

Linear models



- 1 Some friends spent a total of \$5.00 on stickers and badges at a store. A sheet of stickers costs \$0.50 and a badge costs \$1.50.
 - a Write a linear equation that represents the relationship between the number of sheets of stickers, x , and the number of badges, y , the friends bought for \$5.00.
 - b If the friends bought 2 badges, how many sheets of stickers did they buy?
- 2 The cost of a taxi ride is given by $C = 5.5t + 3$, where t is the duration of the trip in minutes.
 - a Calculate the cost of an 11-minute trip.
 - b For every extra minute the trip takes, how much more will the trip cost?
 - c What could the constant value of 3 represent in context?
- 3 A baseball is thrown vertically upward by a baseball player when he is standing on the ground, and the velocity of the baseball, V (in yards per second), after T seconds is given by $V = 120 - 32T$.

- a Complete the table of values:

Time	0	1	2	3	4
Vertical velocity					

- b State the slope of the linear function.
 - c Explain the negative value of V when $T = 4$.
- 4 A carpenter charges a callout fee of \$150 plus \$45 per hour.
 - a Write a linear equation to represent the total amount charged, y , by the carpenter as a function of the number of hours worked, x .
 - b State the slope of the linear function.
 - c Describe what the slope of the line represents in this context.
 - d State the y -intercept.
 - e Describe what the y -intercept represents in this context.
 - f Find the total amount charged by the carpenter for 6 hours of work.



- 5 Mohamad is taking his new Subaru out for a test drive. He had only driven 50 miles in it before and is now driving it down the highway at 75 mi/h .
- a Write a linear equation to represent the total distance, y , that Mohamad had driven in his Subaru as a function of the number of hours, x .
 - b State the slope of the function.
 - c Describe what the slope of the line represents in this context.
 - d State the y -intercept.
 - e Describe what the y -intercept represents in this context.
 - f Find the total distance Mohamad will have driven in his Subaru if his current drive begins at 5:10 pm and finishes at 7:25 pm.

- 6 The table shows the linear relationship between the length of a cellphone call and the cost of the call:

Length of call in minutes (x)	1	2	3
Cost in dollars (y)	7.6	14.4	21.2

- a Write a linear equation to represent the cost of a call, y , as a function of the length of the call, x .
 - b State the slope of the function.
 - c Describe what the slope represents in this context.
 - d State the y -intercept.
 - e Describe what the y -intercept could represent in this context.
 - f Find the cost of a 6-minute call.
- 7 Mario is running a 100 km ultramarathon at an average speed of 9 km/h .
- a Write a linear equation to represent the distance Mario has left to run, y , as a function of the number of hours since the start, x .
 - b State the slope of the function.
 - c Describe what the slope of the line represents in this context.
 - d State the y -intercept.
 - e Describe what the y -intercept represents in this context.
 - f Find the distance Mario will have left to run after 4.5 hours.



- 8 A particular restaurant has a fixed weekly cost of \$1300 and receives an average of \$16 from each customer.
- Write an equation to represent the net profit, y , of the restaurant for the week as a function of the number of customers, x .
 - Find the slope of the function.
 - Describe what the slope of the line represents in context.
 - Find the y -intercept.
 - Describe what the y -intercept represents in context.
 - Find the restaurant's net profit if it has 310 customers for the week.
- 9 A cellphone salesman earned \$600 in a particular week during which he sold 26 phones and \$540 in another week during which he sold 20 phones.
- Write a linear equation to represent the weekly earnings of the salesman, y , as a function of the number of phones sold, x .
 - State the slope of this function.
 - Describe what the slope of the line represents in context.
 - Find the y -intercept.
 - Describe what the y -intercept represents in context.
 - Find how much the salesman will earn in a week during which he sells 36 phones.

- 10 In a study, scientists found that the more someone sleeps, the quicker their reaction time. The table below displays the findings:

Number of hours of sleep (x)	0	1	2	3	4	5
Reaction time in seconds (y)	6	5.8	5.6	5.4	5.2	5

- How much does the reaction time decrease for each extra hour of sleep?
- Write a linear equation relating the number of hours of sleep, x , and the reaction time, y .
- Calculate the reaction time for someone who has slept 4.5 hours.
- Calculate the number of hours someone sleeps if they have a reaction time of 5.5 seconds.



- 11 A diver starts at the surface of the water and begins to descend below the surface at a constant rate. The table below shows the depth of the diver, in yards, over 4 minutes:

Number of minutes passed (x)	0	1	2	3	4
Depth of diver in yards (y)	0	1.4	2.8	4.2	5.6

- a Calculate the increase in depth each minute.
 - b Write a linear equation for the relationship between the number of minutes passed, x , and the depth, y , of the diver.
 - c Calculate the depth of the diver after 6 minutes.
 - d Calculate how long the diver takes to reach 12.6 yd beneath the surface.
- 12 After Mae starts running, her heartbeat increases at a constant rate.

- a Complete the following table:

Number of minutes passed (x)	0	2	4	6	8	10	12
Heart rate (y)	49	55	61	67	73	79	

- b What is the unit change in y for the above table?
 - c Write a linear equation that describes the relationship between the number of minutes passed, x , and Mae's heartbeat, y .
 - d Describe what the slope represents in this context.
- 13 Paul has just purchased a prepaid phone, which he intends to use exclusively for sending text messages, and has purchased some credit along with it to use. After sending 11 text messages, he has \$34.39 of credit remaining and after sending 19 text messages, he has \$30.31 of credit remaining.
- a The relationship between the number of text messages sent and the amount of credit remaining is linear. Determine the slope of the linear function.
 - b Write a linear equation to represent the amount of credit remaining, y , as a function of the number of text messages sent, x .
 - c Describe what the slope of the line represents in this context.
 - d State the value of the y -intercept.
 - e Describe what the y -intercept represents in this context.
 - f Find how much credit Paul will have left after sending 36 text messages.



- 14 A racing car starts the race with 150 L of fuel. When there, it uses fuel at a rate of 5 L per minute.

a Complete the following table of values:

Number of minutes passed (x)	0	5	10	15	20
Amount of fuel left in the tank (y)					

- b Write a linear equation linking the number of minutes passed, x , and the amount of fuel left in the tank, y .
- c Explain the meaning of the slope in this context.
- d How many minutes will it take for the car to run out of fuel?
- 15 The table shows the linear relationship between the number of plastic chairs manufactured and the total manufacturing cost:

Number of plastic chairs (x)	2	4	7
Cost in dollars (y)	135	185	260

- a Write a linear equation to represent the total manufacturing cost, y , as a function of the number of plastic chairs manufactured, x .
- b State the slope of the function.
- c Describe what the slope of the function represents in this context.
- d State the y -intercept.
- e Describe what the y -intercept could represent in this context.
- f Find the total cost of manufacturing 13 plastic chairs.

- 16 The table shows the water level of a well that is being emptied at a constant rate with a pump:



Time in minutes (x)	2	5	8
Water level in meters (y)	26.8	25	23.2

- a Write a linear equation to represent the water level, y , as a function of the minutes passed, x .
 - b State the slope of the function.
 - c Describe what the slope of the function represents in this context.
 - d State the y -intercept.
 - e Describe what the y -intercept represents in this context.
 - f Calculate the water level be after 15 minutes.
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Graphs of linear models

- 17 There are 20 L of water in a rainwater tank. It rains for a period of 24 hours and during this time, the tank fills up at a rate of 8 L per hour.

- a Complete the table of values:

Number of hours passed (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

- b Write a linear equation linking the number of hours passed, x , and the amount of water in the tank, y .
- c Plot the points on a coordinate plane.

18 Consider the points in the following table:



Time in minutes (x)	1	2	3	4	5
Temperature in $^{\circ}\text{F}$ (y)	8	13	18	23	28

- a By how much does the temperature increase each minute?
- b What would the temperature have been at time 0?
- c Write a linear equation that represents the relationship between x and y .
- d Sketch the graph of the linear relationship.

19 A carpenter charges a callout fee of \$90 plus \$30 per hour.

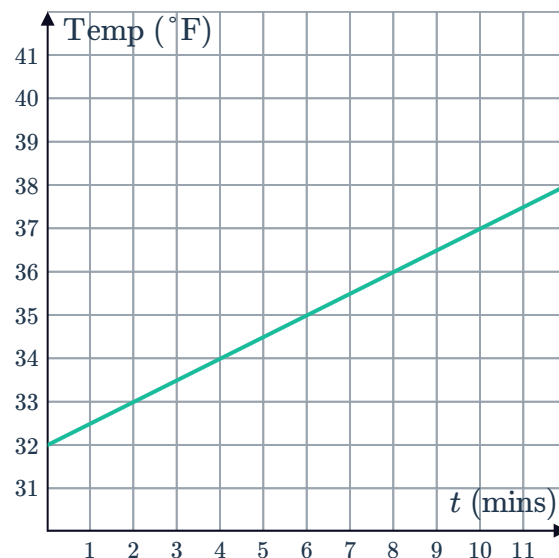
- a Write a linear equation to represent the total fee charged, y , by the carpenter as a function of the number of hours worked, x .
- b Complete the table of values:

Number of hours worked (x)	0	1	2	3	4	4.5	7
Total fee charged (y)							

- c Sketch the graph of the linear relationship.

20 The graph shows the temperature of a room after the heater has been turned on:

- a State the slope of the line.
- b State the y -intercept.
- c Write a linear equation to represent the temperature of the room, y , as a function of time, t .
- d Describe what the slope represents in this context.
- e Describe what the y -intercept represents in this context.
- f Find the temperature of the room after the heater has been turned on for 40 minutes.



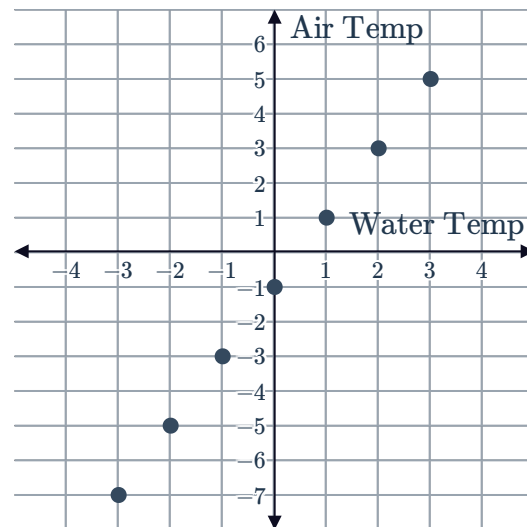
- 21** A ball is rolled down a hill. The table below shows the velocity, V , of the ball after a given number of seconds, t :



Time in seconds (t)	0	1	2	3	4	5
Velocity in m/s (V)	12	13.3	14.6	15.9	17.2	18.5

- Sketch a graph that displays the ball's velocity against time.
- Calculate the slope of the line.
- Describe what the slope represents in this context.
- State the y -intercept of the line.
- Describe what the y -intercept represents in this context.
- Write a linear equation for the line, expressing V in terms of t .
- Now, determine the velocity of the ball after 19 seconds.

- 22** The following graph shows the relationship between water temperatures and surface air temperatures:



- Complete the table of values:

Water temperature ($^{\circ}\text{F}$)	-3	-2	-1	0	1	2	3
Surface air temperature ($^{\circ}\text{F}$)							

- Write a linear equation representing the relationship between the water temperature, x , and the surface air temperature, y .
- Find the surface air temperature when the water temperature is 14°F .
- Find the water temperature when the surface air temperature is 23°F .



23 It starts raining and an empty rainwater tank fills up at a constant rate of 2 gal per hour. By midnight, there are 20 gal of water in the rainwater tank. As it rains, the tank continues to fill up at this rate.

a Complete the table of values:

Number of hours passed since midnight (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

- b** Sketch the graph depicting the situation on a coordinate plane.
- c** Write a linear equation linking the number of hours passed since midnight, x , and the amount of water in the tank, y .
- d** Determine the y -intercept of the line.
- e** At what time prior to midnight was the tank empty?

24 The number of fish in a river is approximated over a five year period. The results are shown in the following table:

Time in years (t)	0	1	2	3	4	5
Number of fish (F)	4800	4600	4400	4200	4000	3800

- a** Sketch a graph corresponding to the given information.
- b** Calculate the slope of the line.
- c** Describe what the slope represents in this context.
- d** State the value of F when the line crosses y -axis.
- e** Determine an equation for the line using the given values.
- f** Now, determine the number of fish remaining in the river after 13 years.
- g** Find the number of years, t , until 2000 fish remain in the river.

25 The graph shows the amount of water remaining in a bucket that was initially full before a hole was made in its side:



- a State the slope of the line.
- b Find the y -intercept.
- c Write a linear equation to represent the amount of water remaining in the bucket, y , as a function of time, x .
- d Describe the meaning of the slope in this context.
- e Describe the meaning of the y -intercept in this context.
- f Find the amount of water remaining in the bucket after 54 minutes.

